Getting Started Tutorial: Mixed-Model Analysis Using the Sleep Study Dataset

Overview

This tutorial demonstrates a complete **Generalized Linear Mixed Model (GLMM)** workflow using the example **sleep study dataset** included with the application. The dataset consists of repeated measurements collected from multiple participants over several days and is structured in **long format**, where each row represents a single observation within a subject.

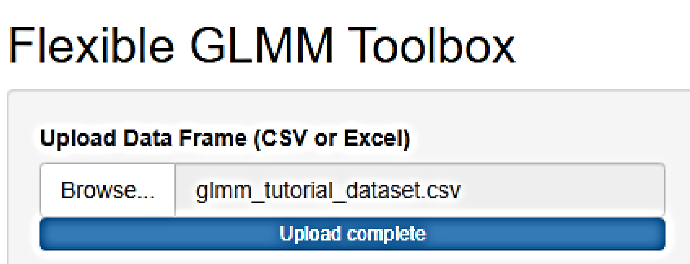
The tutorial covers:

- Data import and preprocessing
- Missing data handling
- Comparison of multiple outlier detection methods
- Distribution fitting
- Model specification using `afex`, `lme4`, and `nlme`
- Model interpretation
- Comparison of different GLMM families and link functions
- Covariance structure modeling for repeated-measures data

Note: For detailed outputs, please refer to the **Tutorial_with_outputs** file.

Step 1: Upload the Dataset

1. Click **Browse** under the *Upload Data Frame (CSV or Excel)* section.
2. Select the **sleep study dataset** (`sleepstudy.csv` or `.xlsx`).
3. Once uploaded, navigate to the **Data** tab and scroll through the dataset.



Expected Outcome

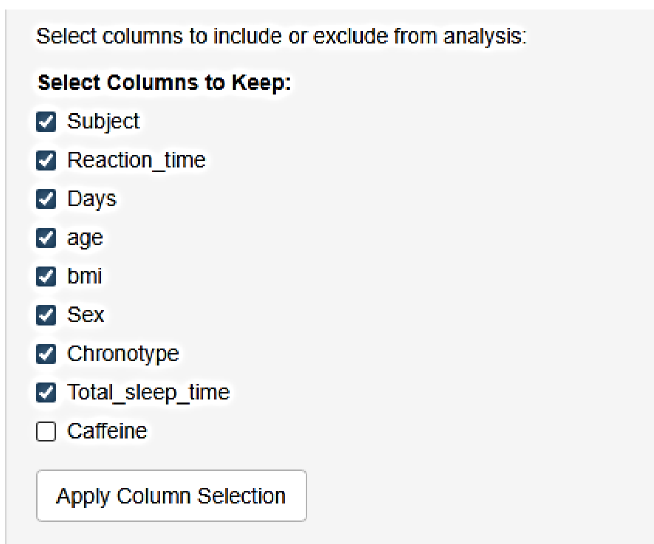
The dataset should appear in **long format**, where:

- Each row represents a repeated observation.
- `Subject` identifies individual participants.
- `Days` represents the repeated measurement/time variable.
- `Reaction_time` is the dependent outcome variable.

Long-format data are required for most mixed-effects modeling approaches.

Step 2: Select Variables for Analysis

1. Under *Select columns to include or exclude from analysis*, select the variables needed for analysis.
2. For this demonstration, **exclude the `caffeine` variable**.
3. Click **Apply Column Selection**.

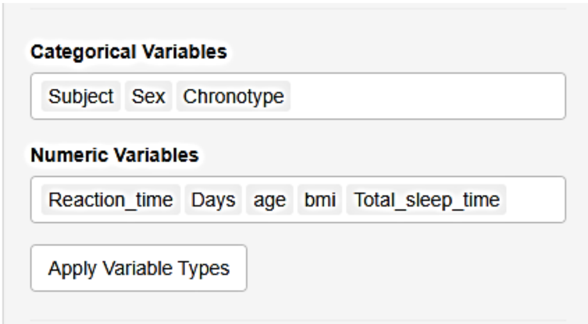


Expected Outcome

The working dataset now contains only the variables used for model fitting and preprocessing.

Step 3: Specify Variable Types

1. Select categorical or factor variables and continuous or numerical variables.
2. Click **Apply Variable Types**.

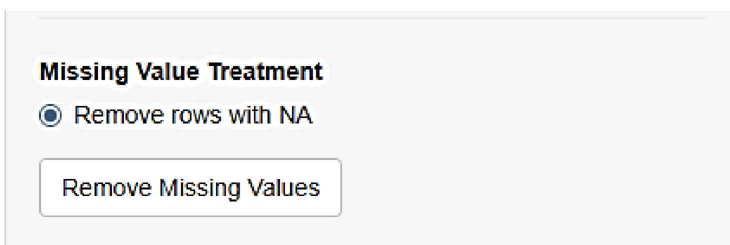


Expected Outcome

The working dataset now classifies the variables either as categorical or continuous for model fitting and preprocessing.

Step 4: Remove Missing Values (Mandatory Step)

1. Select **Remove rows with missing data** (<u>blank/empty cells</u> in the datasheet).
2. Click **Remove Missing Values**.
3. Review the displayed dataset dimensions and missing-value summary.

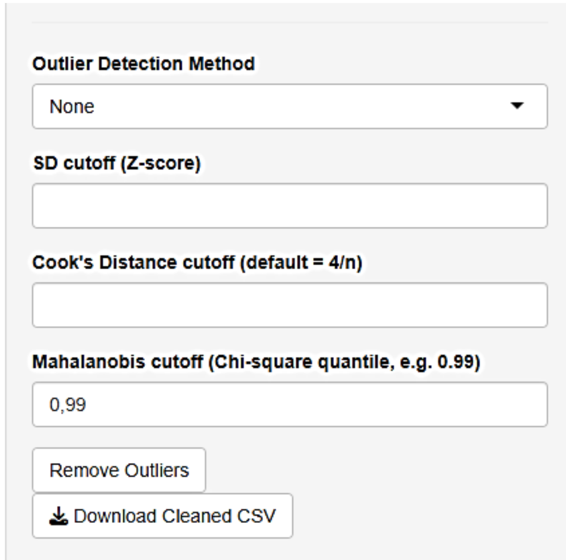


Important

Removing missing values is currently a mandatory preprocessing step before fitting models in FlexibleGLMM.

Step 5: Compare Outlier Detection Methods

FlexibleGLMM provides five approaches for identifying unusual observations. It is recommended to compare multiple methods rather than relying on a single criterion.



5.1 Standard Deviation (Z-score) Method

1. Select **SD-based outlier detection**.
2. Use a cutoff of ≥**3 standard deviations**.
3. Review the identified observations.

A threshold of ±3 standard deviations is commonly used as a conventional rule for identifying outlier values.

5.2 Cook's Distance

1. Select **Cook's Distance**.
2. Leave the cutoff field blank to use the default threshold: [4/n]
3. Alternatively, a cutoff value of **1** may be used as a less stringent criterion.

5.3 Mahalanobis Distance (MD)

1. Select **Mahalanobis Distance**.
2. Use the default chi-square quantile threshold (e.g., **0.99**).

A threshold of **0.95** is also commonly used depending on the desired sensitivity.

5.4 Lookout-Based Outlier Detection

1. Select the **Lookout** method.
2. Review the observations identified as potential outliers.

Lookout provides an alternative approach for detecting unusual observations using leave-one-out kernel density estimates and extreme value theory.

5.5 DHARMa-Based Outlier Detection

1. Select **DHARMa**.
2. Fit a simple preliminary model if required.

Why is a model needed?

DHARMa detects outliers using simulated model residuals; therefore, a fitted model is required before diagnostic plots can be generated.

Step 6: Decide on Outlier Removal

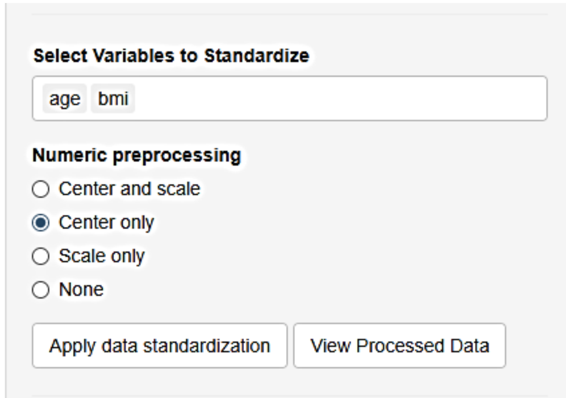
After comparing the results from all five methods:

- Evaluate whether detected observations represent true errors or valid biological variability.
- Do not remove observations automatically.

For this tutorial, based on the consensus of the different methods, **no observations will be removed**.

Step 7: Perform Data Standardization

1. Select the numerical variables to be standardized.
2. Select numerical processing steps to perform.
3. Choose either **None** or **Center only** (if afex is used) for this tutorial.
4. Click **View Processed Data** and check for additional columns with suffix: **_cs** (for center & scale), **_c** (for center only), **_s** (for scale only).



Note

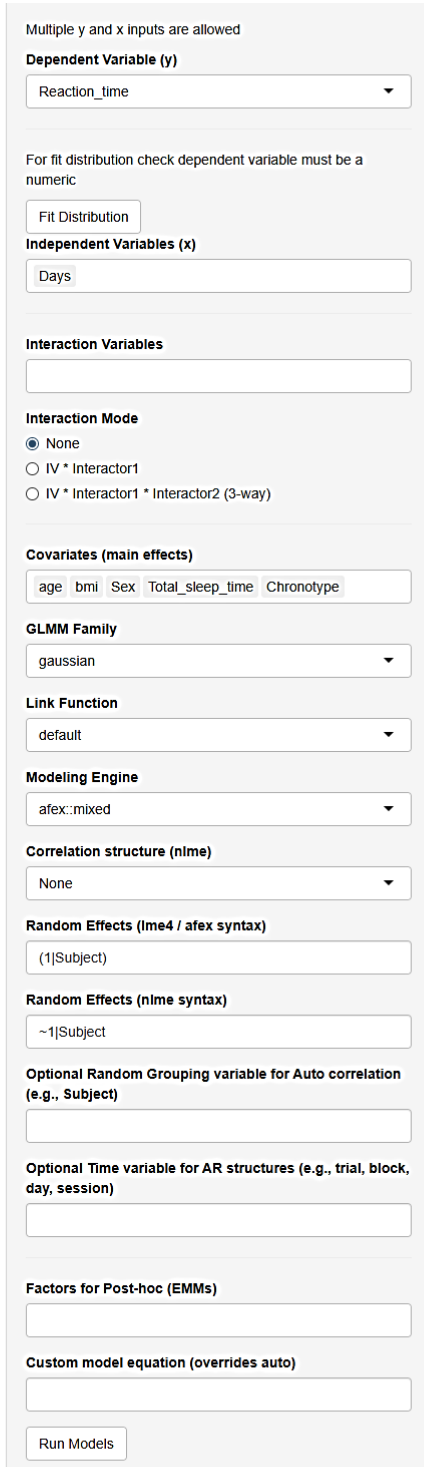
Numerical standardization (centering or scaling) can influence certain distance-based outlier detection methods. Therefore, outlier assessment should be performed on the original scale of the variables whenever possible.

Step 8: Specify the Initial Model

Configure the model as follows:

Option	Selection
Dependent variable	`Reaction_time`
Independent variable	`Days`
Interaction	None
Random effects	`(1|Subject)`
Modeling engine	`afex::mixed`

Click **Run Models**.



Note

Multiple x and y imply that multiple dependent and multiple independent variables can be used to run several models simultaneously.

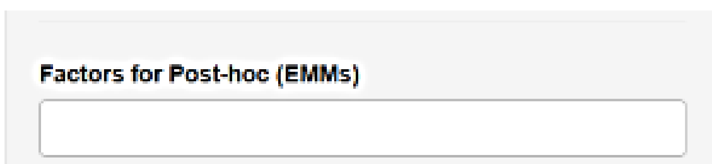
Step 9: Review the Initial Gaussian Model Output

Inspect the following tabs:

- **Model Output** – Fixed and random effect estimates.
- **ANOVA** – Type III tests comparable to SAS PROC MIXED.
- **Summary Table** – Publication-ready model summaries.
- **Performance** – Model fit statistics and explanatory measures.
- **Diagnostics** – Residual behavior, influential observations, and assumption checks.

Step 10: Run Post hoc analysis

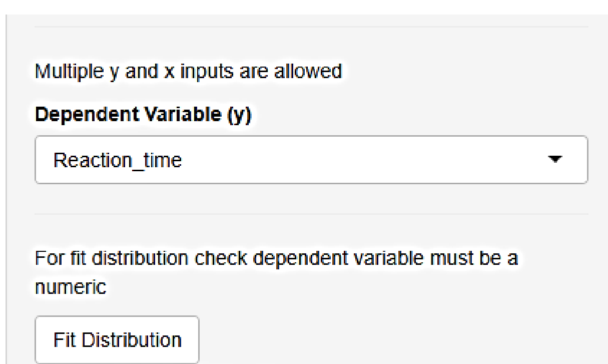
1. Select categorical variables to perform Tukey's post hoc analysis.
2. Choose either **Sex** for this tutorial.



Step 11: Assess the Response Distribution

Return to the **FitDist Output** tab.

Review the distribution fitting results.



Expected Observation

For the sleep study data, the Gamma distribution may provide a better statistical fit than the Gaussian distribution based on the goodness-of-fit criteria.

Step 12: Refit the Model Using a Gamma Distribution

Update the model settings:

Option	Selection
Family	Gamma
Link	Identity

Click **Run Models**.

Why use Identity Link?

The default link for Gamma models is the **inverse link**, which often provides better numerical convergence. However, the **identity link** may be preferred because model coefficients remain directly interpretable on the original outcome scale.

Step 13: Compare Gamma Link Functions

Repeat the Gamma model using the default **inverse link**.

Compare:

- Parameter estimates
- Model fit statistics
- Interpretation of coefficients

This demonstrates how changing the link function can alter model interpretation even when the same response distribution is used.

Step 14: Compare Gaussian Model with a Log Link

Fit another model using:

Option	Selection

Option	Selection
Family	Gaussian
Link	Log

Important Note

A Gaussian model with a log link is **not equivalent to a log-normal model**. The choice of family and link function determines the assumed relationship between the mean and the linear predictor.

Step 15: Fit a Repeated-Measures Model with Covariance Structure

To explicitly model within-subject correlation, configure the following:

Option	Selection
Family	Gaussian
Link	Default
Modeling engine	`nlme`
Correlation structure	Select the desired covariance option
Random effects	Appropriate `nlme` random-effect specification

Click **Run Models**.

Expected Outcome

The `nlme` engine allows explicit modeling of residual correlation structures commonly used in repeated-measures analyses, including structures such as:

- Compound symmetry
- Autoregressive AR(1)
- Other supported covariance models

This functionality closely resembles repeated-measures analyses performed in **SAS PROC MIXED**.

Summary

In this tutorial, you have learned how to:

- Import and inspect long-format repeated-measures data.
- Select relevant variables.
- Handle missing observations.
- Compare five different outlier detection methods.
- Perform Data Standardization.
- Evaluate the most suitable response distribution.
- Fit mixed models using `afex`, `lme4`, and `nlme`.
- Obtain SAS-like Type III ANOVA results.
- Compute model summaries and diagnostics.
- Compare different GLMM families and link functions.
- Model repeated-measures covariance structures using `nlme`.

This example demonstrates the complete end-to-end workflow of FlexibleGLMM, from raw data preprocessing to advanced mixed-model inference and reporting.